



myNRC-OCR: A Hybrid OCR Framework for Handwritten Myanmar NRCs Integrating YOLOv8, Tesseract LSTM, TrOCR, and Post-OCR Correction

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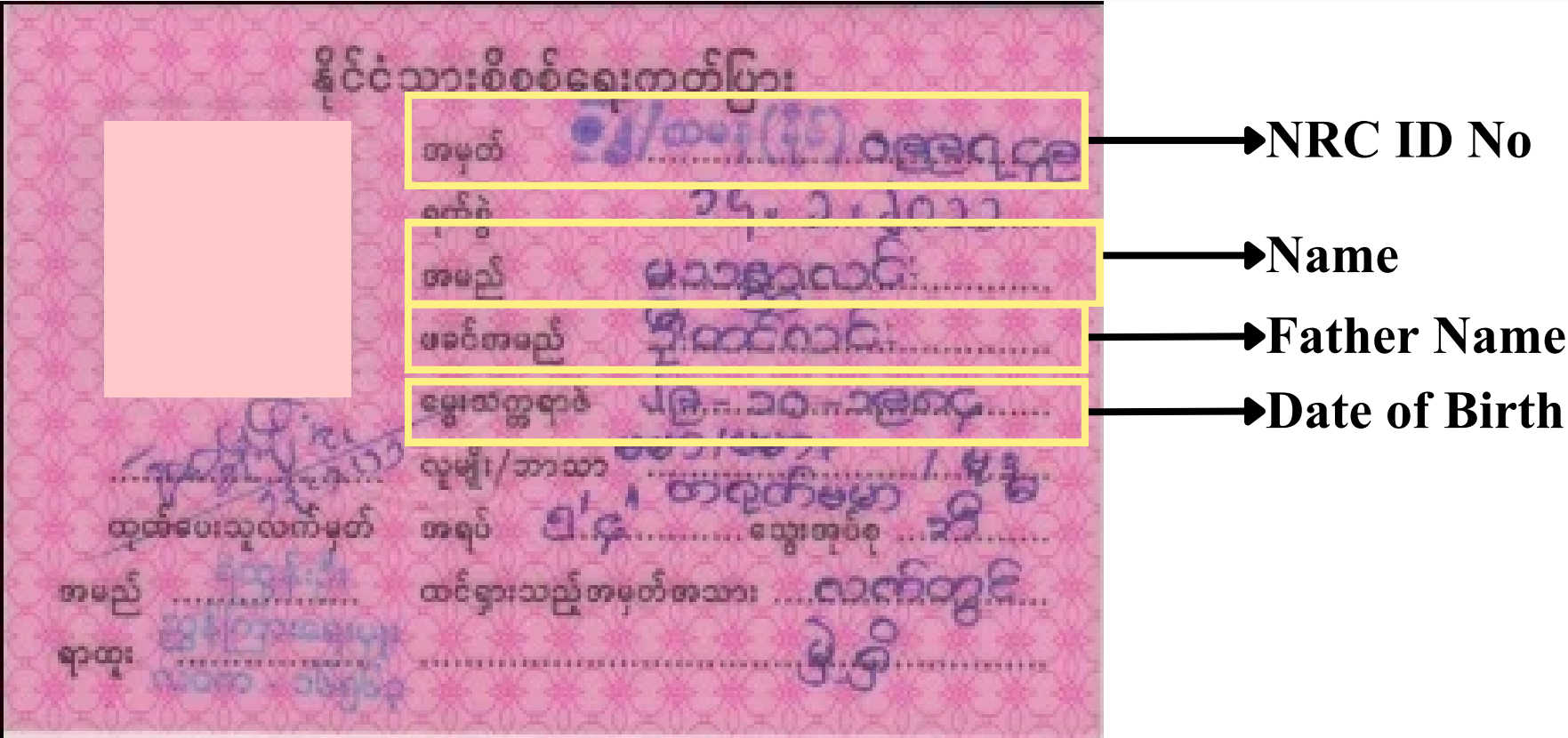
Research Collaboration: University of Information Technology (Myanmar) × Dinger (Industry Partner) × CMKL University (Thailand)

1. INTRODUCTION

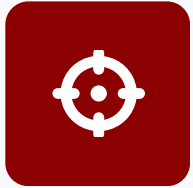
The Myanmar OCR Challenge

- The Big Picture:** Digitizing handwritten documents in low-resource linguistic environments is a major bottleneck for document image analysis.
- Complex Script Geometry:** Myanmar script features circular orthography, non-linear character stacking, and intricate diacritics.
- Phonetic Dependencies:** Stacked consonants create structural variations that traditional OCRs fail to map accurately.
- The Target:** Myanmar National Registration Cards (myNRCs)—a critical document for KYC (Know Your Customer) automation.
- Noisy Backgrounds:** NRCs feature high-entropy, decorative backgrounds that disrupt standard image binarization and segmentation.
- The Goal:** Develop a robust, scalable framework (myNRC-OCR) to automate text extraction from high-variance handwritten Myanmar script.

Fig: Sample of Myanmar NRC Card from ResearchGates on the Internet



Research Contributions



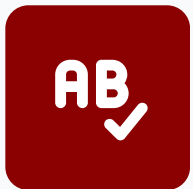
Field Localization

YOLOv8 identifies and crops specific NRC fields (**ID, Name, Father's Name, DOB**) to isolate textual regions from complex document backgrounds.



2. Comparative Recognition Pipeline

A dual-path approach evaluating the stability of **Tesseract LSTM** against the generative flexibility of **Transformer-based TrOCR**. Analysis reveals a shift from segmentation-based failures in LSTMs to generative hallucinations in Transformers.



3. 3-Tier Post-OCR Correction

A **3-tier post-processing module** employing syllable tokenization and fuzzy matching to rectify orthographic errors. Reduces CER from through SymSpell, N-gram ranking, and heuristic fallback.

Dataset Preparation Overview

Data Acquisition

Source: High-resolution NRC images from AWS production environment

Conditions: Diverse real-world noise and lighting variations

Consent: Explicit user consent for system improvement

Privacy: PII encrypted and isolated per regional data protection guidelines

Preprocessing Pipeline

- 1 YOLOv8 Localization:** Isolate Name, Father's Name, DOB, NRC ID regions
- 2 Grayscale Conversion:** Reduce color channel complexity and Save storage
- 3 Otsu Thresholding:** Remove decorative backgrounds, enhance stroke contrast

Dataset Organization

YOLOv8 Detection **369 images**

Total NRC Cards (Each Card has four bboxes: Name, NRC ID, DOB)

Tesseract OCR **31,038 images**

Total images (Name, NRC ID, DOB)

TrOCR Training **31,407 images**

Total images (Name, NRC ID, DOB)

Data Availability

Synthetic sample dataset available on Hugging Face for reproducibility:
huggingface.co/datasets/PyaeLinn/myNRC-OCR_Sample-Dataset

Object Detection Dataset (YOLOv8)

Dataset Distribution

Split	Images	Percentage
Train	258	70%
Valid	74	20%
Test	37	10%
Total	369	100%

Key Insight

While the dataset comprises **only 369 images**, each sample contains **multiple class instances**, providing **sufficient feature density** for learning spatial relationships. Empirical testing confirmed **high localization accuracy** without significant regression.

Detection Classes

Name	
Cardholder's full name field box	
Father Name	
Father's name field box	
NRC No	
National Registration Card number field box	
Date of Birth	
Birth date in Myanmar format filed box	

Tesseract OCR Dataset

The Tesseract dataset uses **a field-specific organization (Name, NRC ID, DOB)** for targeted optimization.

Field-Specific Organization

Field	Images	Characters
Name (Name + Father)	15,602	~374,448
NRC ID	9,460	~236,500
Date of Birth	5,976	~137,448
Total	31,038	~748,396

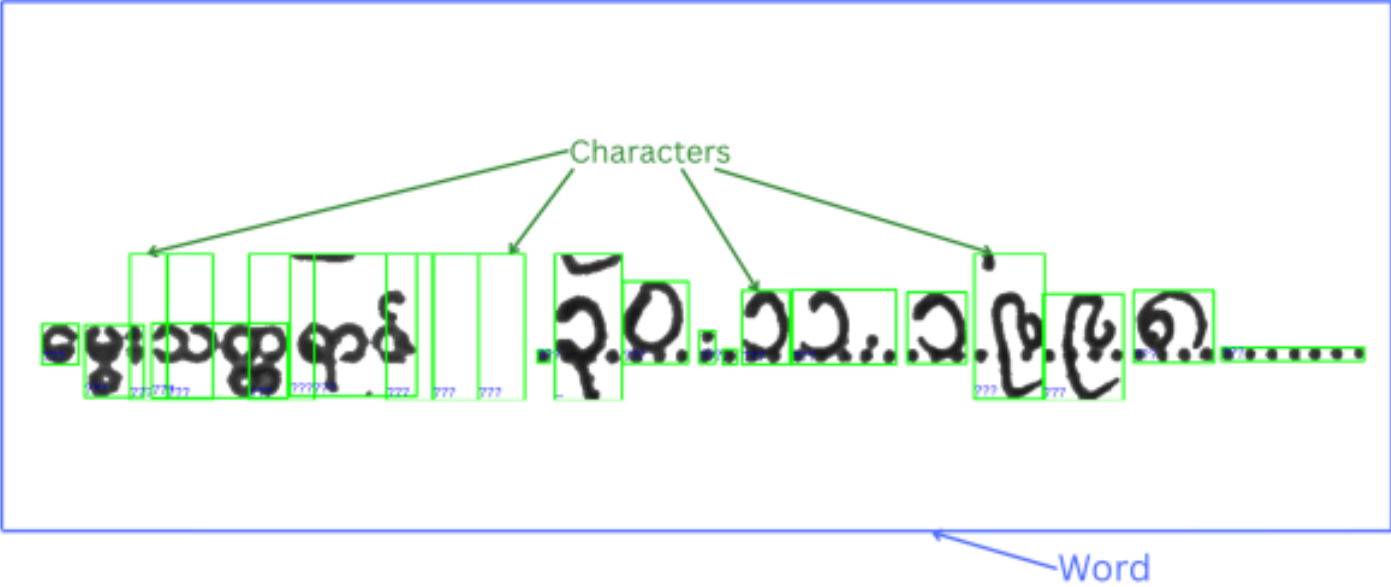


Fig. 1. Character-level Segmentation of the Myanmar phrase "မွေးသက္ကရာဇ် ၃၀.၁၁.၁၉၉၈" (Date of Birth: 30.11.1998)in which Characters in Green and Word in Blue boxes

TrOCR Dataset Distribution

The TrOCR dataset employs a unified image–text pairing strategy partitioned into **training, validation, and testing** subsets for sequence learning.

Unified Image-Text Pairing		
Split	Images	Percentage
Train	21,985	70%
Valid	6,281	20%
Test	3,141	10%
Total	31,407	100%

Three-Stage Pipeline

1
STAGE

Field Localization

YOLOv8 identifies and crops specific NRC fields (ID, Name, Father's Name, DOB) to isolate textual regions from complex document backgrounds.

📍 4 field classes 🖼️ Otsu thresholding

2
STAGE

Hybrid Recognition

Two complementary architectures fine-tuned on localized crops: **Tesseract LSTM** provides robust baseline with CTC, while **TrOCR** handles high-variance handwriting through global attention.

🧠 LSTM baseline 🔄 Transformer ⚖️ Comparative study

3
STAGE

Post-OCR Correction

Final module utilizing **rule-based logic** and **fuzzy matching** to enforce NRC-specific linguistic and structural syntax.

🔤 SymSpell 📈 N-gram ranking 🛡️ Syntax enforcement

Fine-Tuning Tesseract OCR

➡ Convolutional Layers (Visual Feature Extraction)

- 1 **Feature Mapping:** Extracts deep visual features directly from the cleaned handwritten input image.
- 2 **Geometry Capture:** Generates Convolutional Feature Maps to preserve the complex, circular shapes of Myanmar characters.
- 3 **Sequence Conversion:** Transforms visual maps into a linear Feature Sequence to bridge the gap between vision and text.

➡ Recurrent Layers (Deep Bidirectional LSTM)

- 1 **Context Awareness:** Learns the non-linear, sequential, and phonetic dependencies of stacked Myanmar script.
- 2 **Spatial Mapping:** Uses .box files (pixel coordinate maps) compiled into .lstmf files to precisely train the LSTM on handwriting variations.

➡ Transcription Layer (Sequence Decoding)

- 1 **Per-Frame Prediction:** Evaluates frame-by-frame probability distributions output by the LSTM layers.
- 2 **String Generation:** Translates character probabilities into the final Predicted Sequence string.
- 3 **Targeted Fine-Tuning:** Uses mya.traineddata as a baseline, iteratively adjusting the recurrent and transcription layers to master handwritten NRC data.

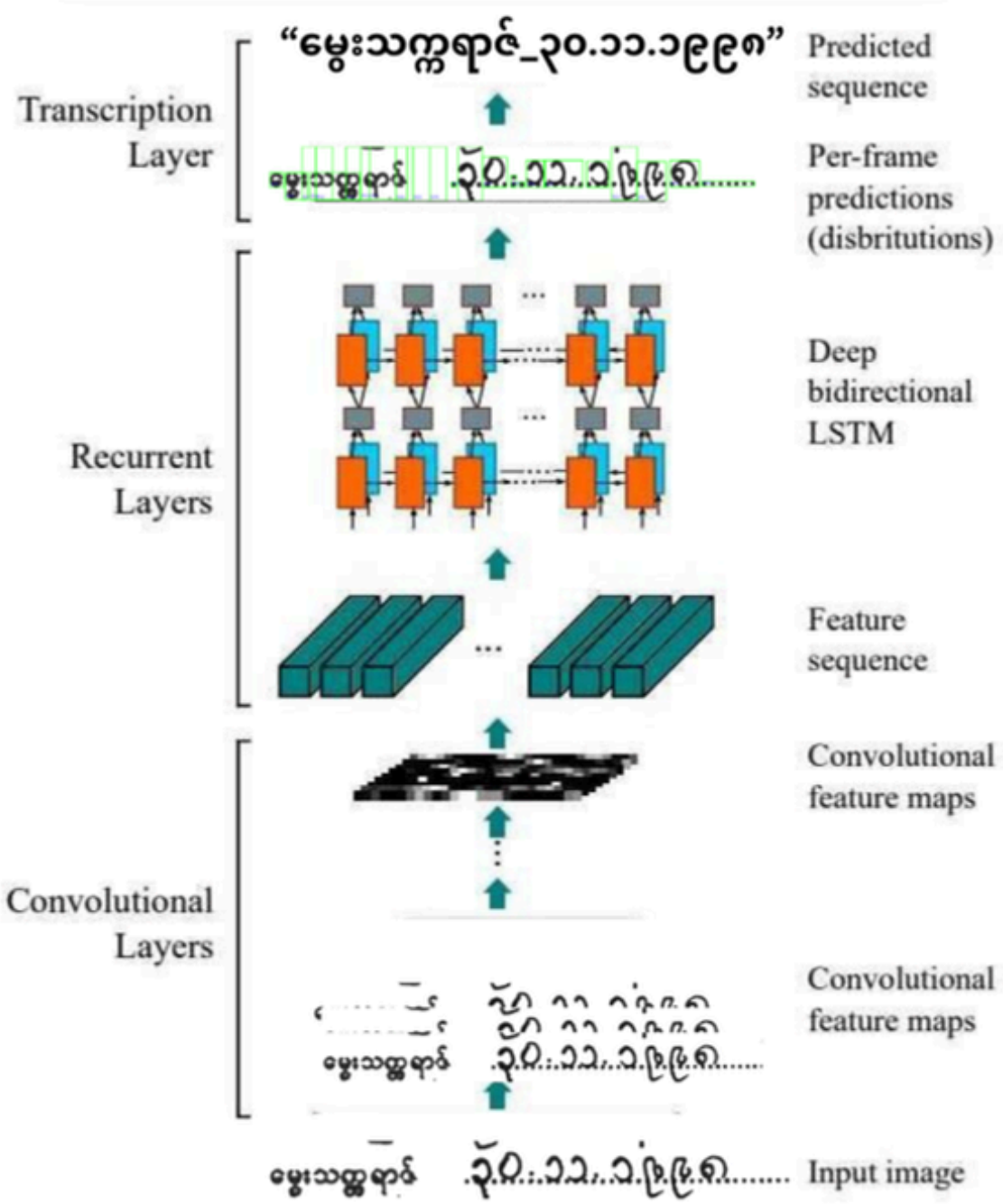


Fig. 2. Architecture of the Fine-Tuned Tesseract OCR Model for Myanmar NRCs (Adapted from: Structure of the Tesseract OCR, ResearchGate Fig. 4, accessed 24 Dec 2025).

Fine-Tuning TrOCR for Myanmar NRC

Dataset & Normalization

- 1 **Linguistic Standardization:** Text normalized to Unicode NFC using unicodedata and regex.
- 2 **Cleaning:** Stripped zero-width spaces (U+200B/U+200C) and removed non-textual artifacts (underscores, numeric suffixes).

Robust Data Augmentation

- 1 **Geometric Changes:** Grid Distortion and Elastic Transformations to simulate localized, irregular handwriting stroke warping.
- 2 **Photometric Noise:** RGB Shifts, Gaussian Noise, and Blur to mimic low-quality, real-world mobile camera captures.
- 3 **Orientation Alignment:** Rotations constrained within $\pm 5^\circ$ to handle real-world scanning alignments.

Hybrid Model Architecture

- 1 **Total Parameters:** 806 million parameters.
- 2 **Vision Encoder:** 12-layer Vision Transformer (ViT) with 768 hidden units to capture visual text features.
- 3 **Text Decoder:** 12-layer SEA-LION BERT decoder with 1,024 hidden units, natively optimized for Southeast Asian languages.
- 4 **Vocabulary Expansion:** Tokenizer expanded to 256,000 tokens for dense Myanmar script coverage.

Fine-Tuning TrOCR for Myanmar NRC

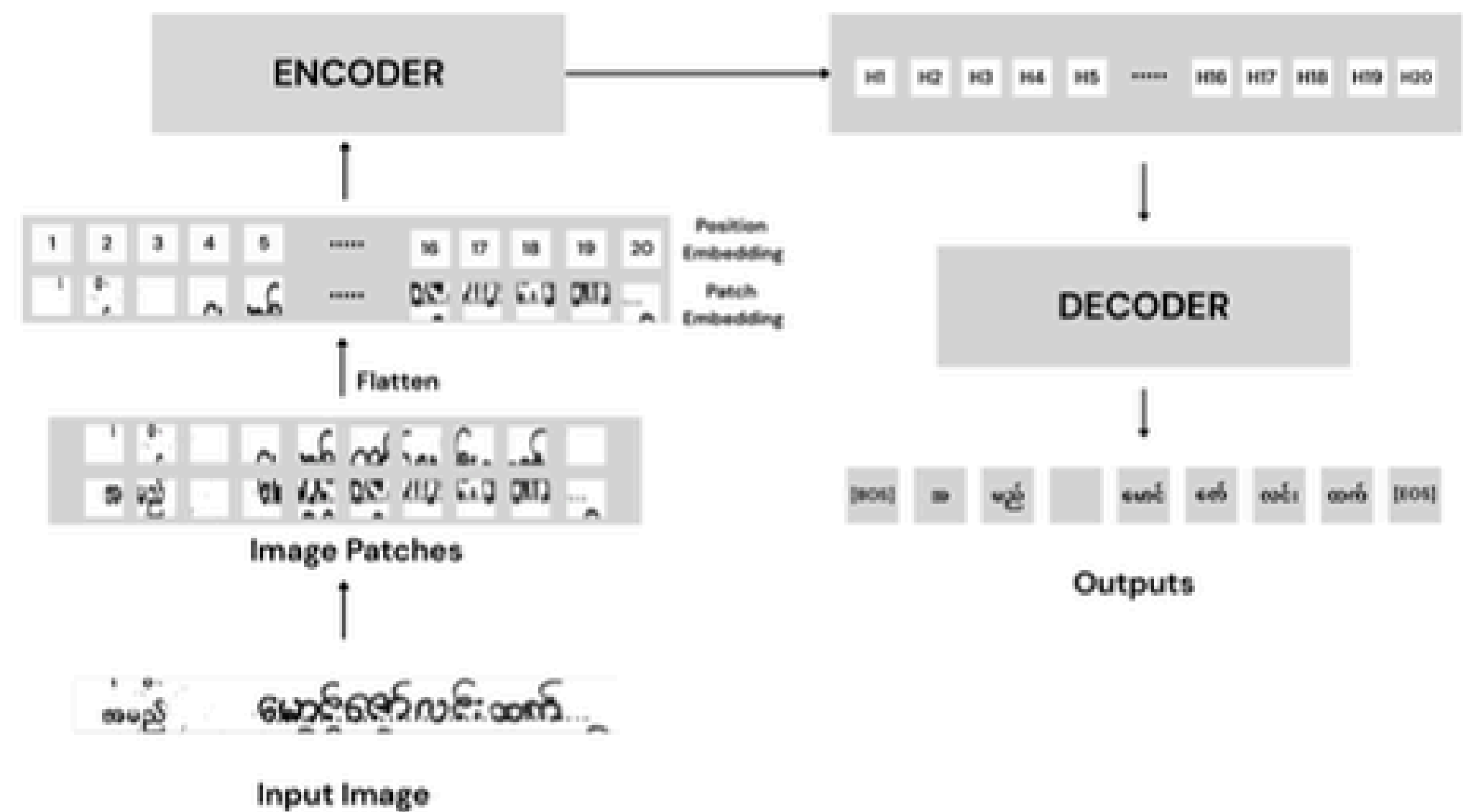


Fig. 3. **Fine-Tuned TrOCR Architecture.** The ViT encoder processes 16×16 patches; the decoder uses cross-attention for autoregressive Myanmar sequence generation.

Post-OCR Correction

Field-Specific Normalization

A post-processing module normalizes and validates OCR outputs against NRC standards to handle handwriting variability and complex scripts.

- 1

NRC ID Validation: Cross-checks township codes against a JSON registry; fixes minor typos using Levenshtein distance ($d=1$). Serial numbers are normalized to a rigid template: အမှတ်_xx/yyy(z)dddddd.
- 2

DOB Validation: Converts script digits to ASCII, clamps invalid dates, and reconstructs into a standard Myanmar script format: မွေးသက္ကရာဇ်_dd.mm.yyyy.

3-Tier Hybrid Name Correction

Names are first segmented into syllables using sylbreak.py, then funneled by N-gram confidence:

- 1

Tier 1 (High Confidence > 0.8): Fast SymSpell dictionary lookup to prevent over-correction.
- 2

Tier 2 (0.5–0.8): SymSpell candidates ranked by N-gram frequency.
- 3

Tier 3 (Low Confidence < 0.5): Applies aggressive sub-word fuzzy matching and structural constraints to reconstruct heavily corrupted or hallucinated syllables.

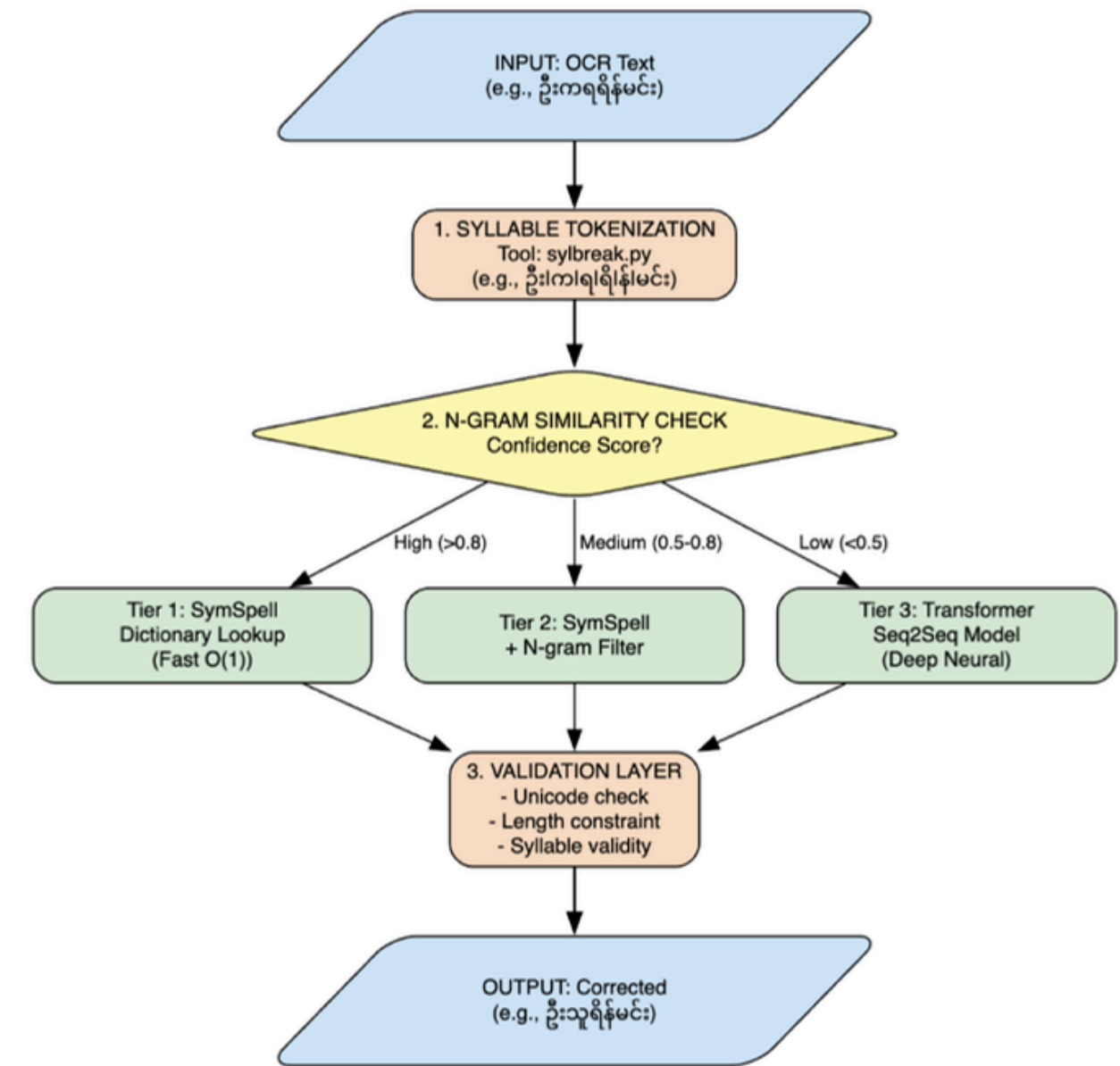


Fig. 4. Overview of the 3-Tier Hybrid Post-Processing Pipeline. The system routes OCR outputs through SymSpell or a Transformer Seq2Seq model based on confidence scores.

Optical Model Training with Tesseract

🔧 Training Configuration

Targeted Learning: Separate, specialized models were trained for individual fields rather than a single monolithic model.

📊 Trained Models:

- 👤 **Name:** Name + Father's Name combined
- 📄 **NRC ID:** Digits and structure
- 📅 **DOB:** Numeric-dominant format

📊 Training Stability

- ✅ **Checkpoint Monitoring:** Employed checkpointed iterative training to track real-time model convergence.
- ✅ **Overfitting Prevention:** Intentionally restricted training bounds to stop the engine from memorizing individual handwriting styles, ensuring scalable generalization.

Training Iterations by Field

Name Model 9,200

High stroke variability requires more iterations

NRC ID Model 8,191

Structured format with digits and patterns

DOB Model 6,318

Numeric-dominant, simpler patterns

Training Infrastructure

Environment:	CPU-only mode
Instance Type:	Compute-optimized
Advantage:	High RAM for glyph caching

Optical Model Training with TrOCR

Hyperparameters

Epochs	30
Learning Rate	1e-5
Weight Decay	0.01
Batch Size (per device)	4
Warmup Ratio	10%
LR Schedule	
Cosine Annealing	

Training Dynamics

Convergence Epoch

18 / 30

Model achieved stable convergence at epoch 18, with diminishing returns thereafter.

Training Optimizations

-  Hardware

NVIDIA L4 GPU (24GB VRAM), Google Colab Pro+
-  FP16 Mixed Precision

Reduces memory footprint, accelerates training
-  Gradient Checkpointing

Trade computation for memory—enable larger batch sizes
-  Gradient Accumulation

Step size of 4—effective batch size 4× larger

Implicit Layout Mastery

-  Segmentation-Free Attention:

Bypasses character segmentation entirely by using an attention mechanism to implicitly map complex vertical stacking and irregular spacing, eliminating the primary failure points of traditional engines like Tesseract.

Tesseract Error Analysis

⚠ Failure Mode Categories

📄 Prefix & Structural Bias

Strong memorization of frequent patterns leads to duplicated or fragmented prefixes

Example: ဖခင်အမည် → အမည်_ဖခင်အမည်_မံ (Case - 2, 5, 6)

⚠ Orthographic & Diacritic Errors

Struggles with complex stacking and diacritics and inserts extra marks or omits characters

Example: အဗ္ဗုဒ္ဓဆတ္တိရ် → အမည်_ (Case - 1, 3, 4)

👁 Visual Ambiguity

Handwritten variations cause misclassification of visually similar characters

Example: မ vs ဗ confusion (Case - 7)

Representative Error Examples

Case	Ground Truth	OCR Error
1	ဦးအောင်ဌေးဦး	ဦးအောင်းဌေး
2	ဖခင်အမည်_စောဘေဆင်း	အမည်_ဖခင်အမည်_မံ
3	အမည်_အဗ္ဗုဒ္ဓဆတ္တိရ်	အမည်_
4	မွေးသက္ကရာဇ်_၁၁.၇.၁၉၉၄	မွေးသက်_၁၁.၇.၁၉၉၄
5	မွေးသက္ကရာဇ်_၁၇.၁၁.၂၀၀၁	မွေးသက္ကရာ မွေးသက္က_၂၀၀၁
6	အမှတ်_၈-၈လန...	အမှတ်_၈- အမှတ်တိုင်...
7	မကန(နိုင်)	ဗကန(နိုင်)

Root Cause Analysis

LSTM sequence instability combined with Myanmar script complexity leads to structural bias, diacritic mishandling, and visual misclassification.

TrOCR Error Analysis

✔ Background Robustness

Unlike Tesseract's susceptibility to **holographic artifacts**, TrOCR's **attention mechanism** isolates **ink density** from **non-semantic noise**. Maintains **high invariance** to **circular NRC patterns** that traditional engines misidentify as characters.

Key Advantage:

- ✔ Attention weights focus on **text regions**
- ✔ Ignores decorative **background patterns**
- ✔ Robust to **low-contrast handwriting**

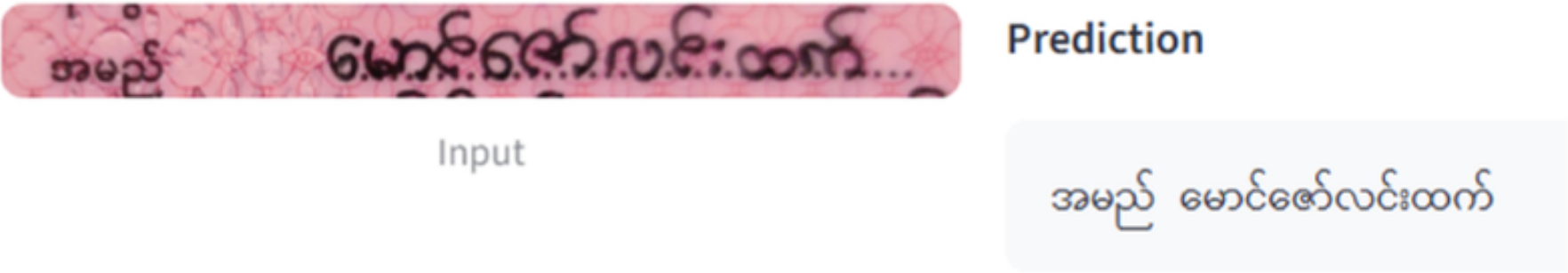


Fig. 5. TrOCR robustness to background artifacts.

! Seq2Seq Hallucinations

TrOCR's limitations stem from prioritizing contextual probability over visual evidence in low-resource settings:

✂ Sequence Truncation

Performance **degrades** on long or compound names (**>5 syllables**) due to positional embedding constraints

✎ Auto-Completion Bias

Decoder may "**hallucinate**" common N-grams (e.g., frequent names) over **faint visual evidence**

AB, Orthographic Forcing

Model occasionally inserts **omitted tone marks** (, ◌◌) to enforce standard script rules, even if absent in handwriting

Post-OCR Correction Experiments

Statistical Correction

SymSpell Integration: Max edit distance 4 for fast candidate generation

N-gram Configuration Testing:

Bigrams (N=2)	Less granular
Trigrams (N=3)	✓ Optimal
4-grams (N=4)	Overly sparse

Result: Trigrams provide superior contextual balance for Myanmar syllable disambiguation.

Heuristic Fallback Tier (Tier 3)

Trigger Condition: N-gram confidence < 0.5 (Tier 3)

- Aggressive Alignment:** Executes Levenshtein-based distance alignment to force synchronization with a curated dictionary.
- Domain-Specific Dictionary:** Uses a valid Myanmar syllable dictionary derived directly from real NRC nomenclature to completely block linguistically impossible strings.
- Efficiency Focus:** Eliminates orthographically invalid sequences while maintaining a fraction of the computational overhead required by a secondary neural network pass.

Evaluation Metrics

Tesseract LSTM Evaluation Method

Character Error Rate (CER) and **Word Error Rate (WER)** calculate error percentage relative to ground truth length N:

$$\text{CER/WER} = (S + D + I) / N \times 100$$

Where S=substitutions, D=deletions, I=insertions

CER: Primary metric for training optimization

WER: Assesses full-field extraction accuracy

TrOCR Evaluation Method

CHRF++ better captures Myanmar script's morphological complexity:

$$\text{CHRF++} = (1+\beta^2)(\text{chr_p} \times \text{chr_r}) / (\beta^2 \cdot \text{chr_p} + \text{chr_r})$$

Balances Character N-gram precision (chr_p) and Character N-gram recall (chr_r)

CHRF++: CHaRacter n-gram F-score plus word n-grams

β: balances the importance of precision and recall.

Post-OCR Correction Evaluation Method

Post-OCR module impact measured by absolute error reduction:

$$\Delta\text{Error} = \text{Error_raw} - \text{Error_corrected}$$

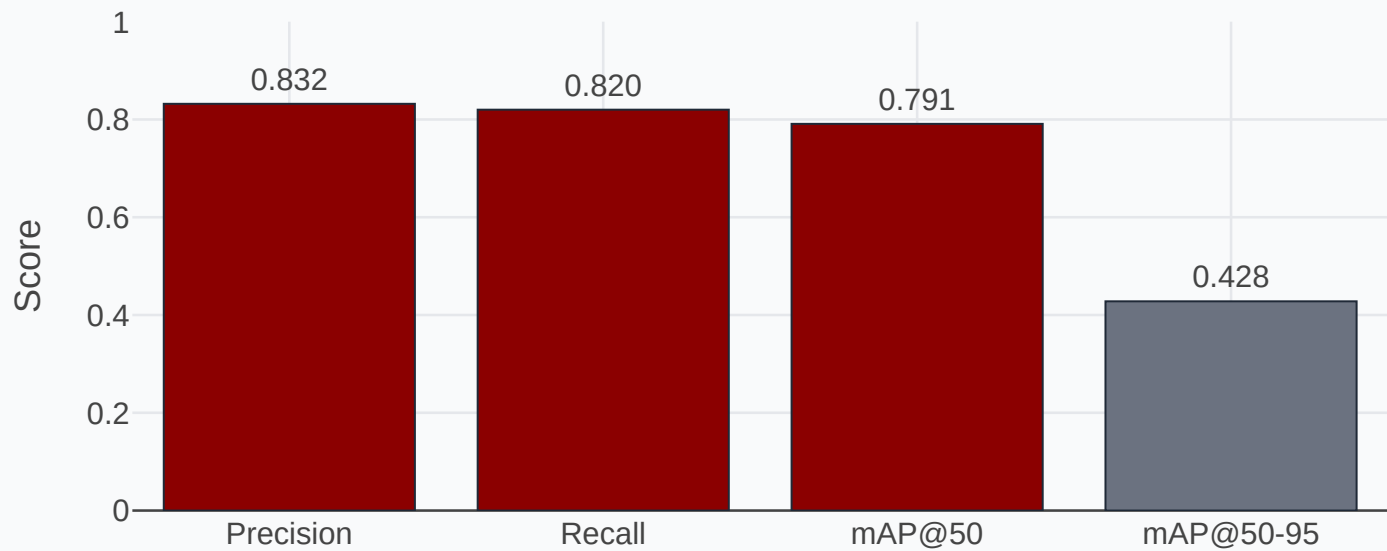
Positive Δ validates normalization and structural enforcement logic

Field Localization (YOLOv8)

Performance Metrics

Precision	0.832
Recall	0.820
mAP@50	0.791
mAP@50-95	0.428
Train Box Loss	1.251
Val Box Loss	1.487

Performance Visualization



Key Findings

- ✔ **Robust Localization:** High precision (0.832) ensures reliable field detection
- ✔ **Good Recall:** 0.820 recall captures most target fields
- ✔ **Background Handling:** Effective despite complex decorative patterns

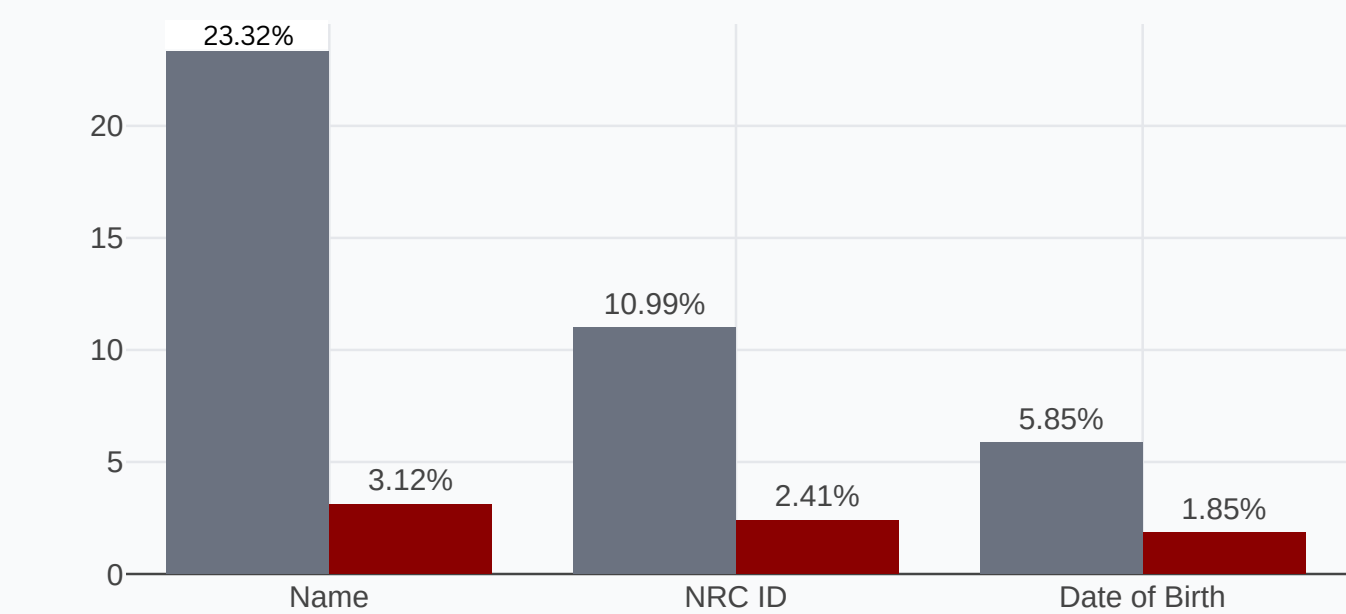
Conclusion: YOLOv8 provides reliable crop generation for downstream OCR, with high recall ensuring minimal missed fields.

OCR Comparison: Tesseract vs TrOCR

Character Error Rate (CER) by Field

Field	Tesseract	TrOCR	Improvement
Name	23.32%	3.12%	20.20%
NRC ID	10.99%	2.41%	8.58%
Date of Birth	5.85%	1.85%	4.00%
Average	13.39%	2.46%	10.93%

CER Comparison Visualization



Comparison Key Points

- ✔ **Accuracy Leap:** TrOCR achieves an average Character Error Rate (**2.46% CER**) compared to **Tesseract's 13.39% CER**.
- ✔ **TrOCR Attention:** isolates ink density from non-semantic noise, maintaining **high invariance to background noises of circular NRC patterns**.
- ✔ **Segmentation Errors:** TrOCR: **Eliminates segmentation errors** entirely unlike Tesseract

Key Insights

- Largest Gain: Name Field**
+20.2% improvement—TrOCR excels at high-variance handwriting
- Consistent Improvement**
All fields show significant CER reduction with TrOCR
- Trade-off Consideration**
Tesseract offers **lower latency**; TrOCR provides **accuracy** for KYC

Post-OCR Correction Evaluation

Ablation Study Results

Step-by-step CER tracking across validation set to quantify each tier's contribution:

Baseline (Raw TrOCR)

The starting point after neural sequence-to-sequence recognition.

2.46%

↓

Tier 1: SymSpell Lookup

Efficiently fixes simple typographical and dictionary misspellings.

2.13%

↓

Tier 2: N-gram Filtering

Uses contextual trigrams to resolve highly similar Myanmar syllables.

1.55%

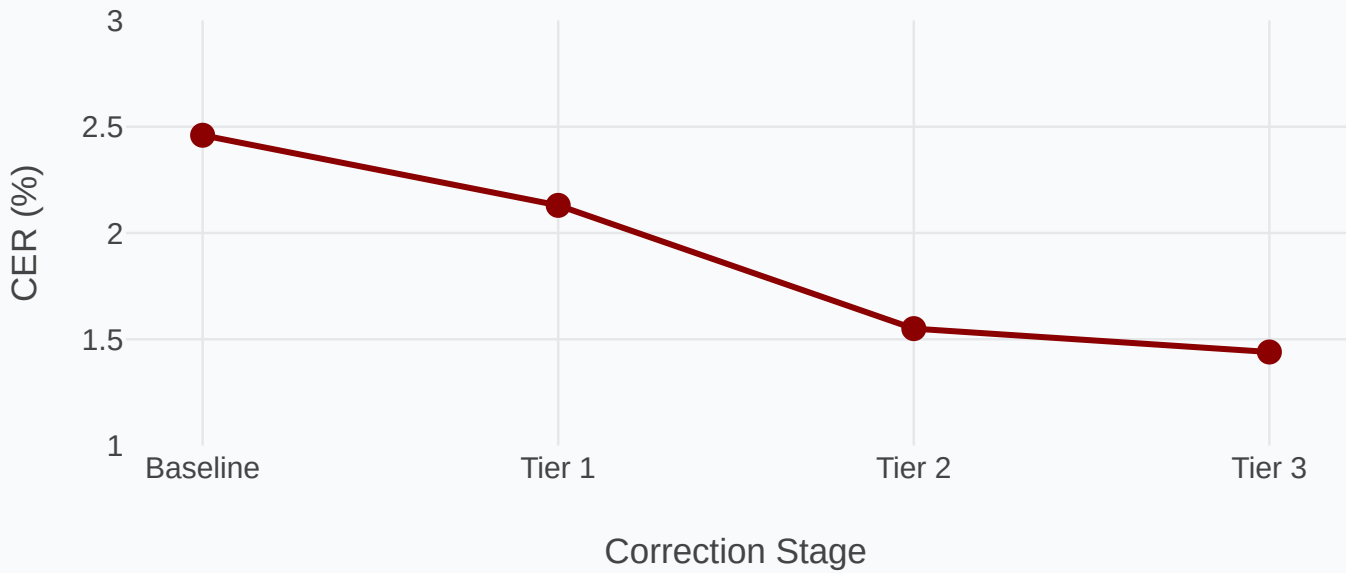
↓

Tier 3: Heuristic Fallback

Reconstructs heavily corrupted or hallucinated text using strict NRC dictionary constraints.

1.44%

CER Reduction Visualization



Progressive CER reduction validates the necessity of the tiered approach.

Overall Improvement

41.5%

Relative CER Reduction

From 2.46% to 1.44%

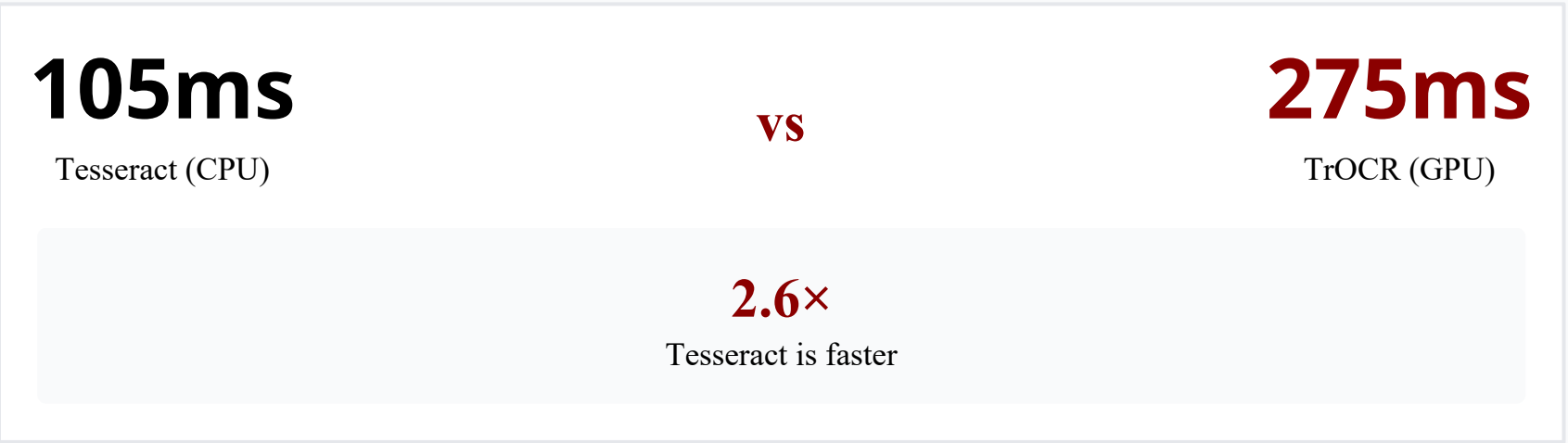
Inference Latency Analysis

On an NVIDIA L4 GPU, the full **TrOCR pipeline averages 275ms per card**, whereas **Tesseract (CPU) achieves 105ms**. While **Tesseract** is approximately **2.6× faster**, the **275ms latency of TrOCR** remains well within the acceptable threshold for automated verification, justifying the trade-off for its superior accuracy.

Latency Breakdown (per NRC Card)

Component	Tesseract (CPU)	TrOCR (GPU)
Field Localization (YOLOv8)	15ms	15ms
OCR Recognition	80ms	250ms
Post-Processing	10ms	10ms
Total Pipeline	105ms	275ms

Speed Comparison



Conclusion & Future Directions

✔ Key Achievements

1. End-to-End Pipeline

Complete OCR system for Myanmar NRC text extraction combining YOLOv8, Tesseract, TrOCR, and post-correction

2. Superior Accuracy

TrOCR achieves **2.46% CER** vs **13.39% for Tesseract**—**10.93%** improvement

3. Post-Processing Efficacy

3-tier correction reduces CER to **1.44%**—**41.5% relative improvement**

4. Production Ready

275ms latency suitable for real-time KYC verification

💡 Research Contributions

- > First comprehensive Myanmar NRC OCR framework
- > Comparative analysis of LSTM vs Transformer architectures
- > 3-tier linguistic refinement for low-resource scripts
- > Methodological blueprint for Southeast Asian script OCR

🔑 Future Enhancements

📄 Dataset Expansion

Increase dataset size for all NRC fields to improve model generalization

🔄 Continuous Fine-Tuning

Refine-tuning with the new updated dataset to improve model robustness

🔧 Enhanced Post-OCR

Combine rule-based and learning-based correction techniques

⚙️ MLOps Deployment

Adopt MLOps pipelines for scalable fintech KYC integration

Impact Vision

Enable seamless digitization of Myanmar identity documents, supporting financial inclusion through automated KYC verification while preserving linguistic heritage.

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Thank You!

Any Questions ?